

Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching

Code:

```
geya@LenovoL480: ~/OS/Skil × + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int pid;

    pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    if (pid == 0) {
        // Child process
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());

        execlp("ls", "ls", "-l", NULL);

        // Runs only if exec fails
        printf("Error: Could not execute command\n");
        return 1;
    }
}
```

```

    else {
        // Parent process
        printf("Parent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}

```

Output:

```

geya@LenovoL480:~/OS/Skill/Week 3$ gcc task4.c -o task4
geya@LenovoL480:~/OS/Skill/Week 3$ ./task4
Parent Process
Parent PID: 1058
Child PID: 1059
Child Process
Child PID: 1059
total 80
-rwxr-xr-x 1 geya geya 16128 Aug 22 04:01 task1
-rw-r--r-- 1 geya geya  617 Aug 22 04:01 task1.c
-rwxr-xr-x 1 geya geya 16256 Aug 22 04:06 task2
-rw-r--r-- 1 geya geya  906 Aug 22 04:05 task2.c
-rwxr-xr-x 1 geya geya 16128 Aug 22 04:10 task3
-rw-r--r-- 1 geya geya  787 Aug 22 04:10 task3.c
-rwxr-xr-x 1 geya geya 16160 Aug 22 04:17 task4
-rw-r--r-- 1 geya geya  717 Aug 22 04:17 task4.c
Child process completed.
geya@LenovoL480:~/OS/Skill/Week 3$ |

```